MTH6134 Group Project
Week 5: Log-linear Models

Statistical Modelling II

1 Introduction

This analysis investigates whether home advantage and opponent league position are associated with the number of goals scored by Ashford United during the fictional 2023/24 Midland Premier League season. The dataset contains 46 matches, comprising 23 home and 23 away fixtures.

As goals scored is a non-negative count variable, Poisson regression with a logarithmic link was selected. This allows the expected number of goals to depend on match characteristics while ensuring that fitted values remain positive.

2 Exploratory Analysis

Ashford scored 60 goals across 46 matches, averaging 1.30 goals per match. The mean number of goals was 1.74 at home compared with 0.87 away, suggesting a possible home advantage.

Opponent rank was included to investigate whether Ashford scored more goals against teams positioned lower in the league. A higher numerical rank represents a lower league position.

3 Statistical Model

Let Y_i denote the number of goals scored in match i. We assume

$$Y_i \sim \text{Poisson}(\mu_i),$$

where μ_i is the expected number of goals scored.

The model uses the logarithmic link:

$$\log(\mu_i) = \beta_0 + \beta_1 H_i + \beta_2 R_i,$$

where $H_i = 1$ for home matches and 0 for away matches, and R_i is the opponent's league rank.

The fitted model was

$$\log(\hat{\mu}_i) = -0.8988 + 0.6932 H_i + 0.05435 R_i.$$

Exponentiating a coefficient gives the multiplicative change in expected goals associated with a one-unit increase in its predictor, holding the other predictor fixed.

4 Model Selection

Likelihood-ratio tests were used to assess whether each explanatory variable improved model fit. The null hypothesis in each test was that the additional coefficient was zero.

Adding opponent rank to the home-only model significantly improved fit ($\chi^2_1 = 8.263$, $p = 0.0040$). Similarly, adding home advantage to the rank-only model improved fit ($\chi^2_1 = 6.796$, $p = 0.0091$). Both null hypotheses were therefore rejected at the 5% significance level.

The model containing both predictors had the lowest Akaike Information Criterion (AIC), supporting its selection over the two simpler alternatives.

Table 1: Comparison of candidate Poisson models

Model	AIC
Home/away only	139.696
Opponent rank only	138.229
Home/away + opponent rank	**133.433**

5 Results and Interpretation

Table 2 reports estimated rate ratios and their 95% confidence intervals.

Table 2: Estimated rate ratios from the Poisson model

Predictor	Rate ratio	95% CI	p-value
Home vs away	2.000	(1.169, 3.421)	0.0114
Opponent rank	1.056	(1.016, 1.097)	0.0051

Holding opponent rank constant, Ashford was estimated to score twice as many goals at home as away. The 95% confidence interval suggests that the expected home-to-away goal ratio lies between 1.169 and 3.421.

Each one-position increase in the opponent's numerical league rank was associated with approximately 5.6% more expected goals, holding venue constant. Both confidence intervals exclude one, providing evidence that these predictors are associated with goals scored.

6 Overdispersion and Model Fit

The fitted model had a residual deviance of 49.785 on 43 degrees of freedom. The Pearson dispersion estimate was

$$\hat{\phi} = \frac{\sum_i r_{P,i}^2}{43} = 0.990,$$

where $r_{P,i}$ is the Pearson residual for match i, and $\hat{\phi}$ is the estimated dispersion parameter.

Since the dispersion estimate is close to one, there is no evidence of substantial overdispersion. Thus, a quasi-Poisson correction does not appear necessary on this basis. Further residual and influence diagnostics are considered in the Week 4 analysis.

7 Conclusion

Both home advantage and opponent rank were significant predictors of goals scored, with the selected Poisson model achieving the lowest AIC among the models considered. The results suggest that Ashford scored more goals at home and against lower-ranked opponents. These are observational associations, and the conclusions remain subject to further diagnostic assessment.

Individual Contributions

Aziz Choudhury

- **Week 5:** Developed the Poisson regression model to investigate the effects of home advantage and opponent rank on goals scored. Conducted model selection using likelihood-ratio tests and AIC, interpreted rate ratios and confidence intervals, and assessed overdispersion.
- **Week 4:** Responsible for goodness-of-fit testing and residual analysis of the Poisson model.

A R Code: Week 5

```r
# Load the match data
matches <- read.csv("/ashford_matches.csv")

# Inspect the dataset
head(matches)
str(matches)
summary(matches)

# Set Away as the reference category
matches$home_away <- factor(
  matches$home_away,
  levels = c("Away", "Home")
)

# Compare mean goals scored by venue
aggregate(
  goals_scored ~ home_away,
  data = matches,
  FUN = mean
)

# Fit the Poisson regression model
model1 <- glm(
  goals_scored ~ home_away + opponent_rank,
  family = poisson(link = "log"),
  data = matches
)

# Examine coefficients and significance
summary(model1)

# Convert coefficients to rate ratios
exp(coef(model1))

# Examine goodness-of-fit statistics
deviance(model1)
df.residual(model1)

# Calculate Pearson dispersion estimate
sum(residuals(model1, type = "pearson")^2) /
  df.residual(model1)

# Fit alternative models for comparison
model_home <- glm(
  goals_scored ~ home_away,
  family = poisson,
  data = matches
)

model_rank <- glm(
  goals_scored ~ opponent_rank,
  family = poisson,
  data = matches
)

# Likelihood-ratio tests
anova(model_home, model1, test = "Chisq")
anova(model_rank, model1, test = "Chisq")

# Compare models using AIC
AIC(model_home, model_rank, model1)

# Calculate rate ratios and Wald confidence intervals
b <- coef(model1)
se <- sqrt(diag(vcov(model1)))

round(cbind(
  Rate_Ratio = exp(b),
  Lower = exp(b - 1.96 * se),
  Upper = exp(b + 1.96 * se)
), 3)
```